



UNIVERSITY EXAMINATIONS: 2016/2017
EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN INFORMATION TECHNOLOGY
BIT2209 RESEARCH SKILLS AND DESIGN
FULL TIME/PART TIME

SPECIAL/SUPPLEMENTARY EXAMINATION

DATE: JULY, 2017

TIME: 2 HOURS

INSTRUCTIONS: Answer **Question One** & ANY OTHER **TWO** questions.

QUESTION ONE (30 MARKS)

- a) Describe what is hypothesis (2 Marks)
- b) Briefly explain two types of hypothesis. Give an example for each case (4 Marks)
- c) Consider the following research topics
- i) Use of ICT in supporting education in KCA University
 - ii) Internet Access in KCA University
 - iii) Mobile phones access in KCA University
- Explain the interplay between the following Research components
- Problem statements
 - Proposed solution
 - Research objectives
- (6 Marks).
- d) Distinguish between secondary and primary data. Give examples to illustrate your answer (4 Marks)
- (e) State and explain Four types of sampling methods (4 Marks)
- (f) Describe the meaning of the term “research” (2 Marks)
- (g) Discuss the difference between the following research concepts. Use one example to illustrate each concept
- i) Sample and population (4 Marks)
 - ii) qualitative data and quantitative data (4 Marks)

QUESTION TWO (20 MARKS)

(a) Explain the following Research Concepts (10 Marks)

- i) Research proposal
- ii) Research Process
- iii) Research Skills
- iv) Research question
- v) literature review

(b). Outline a typical Structure of a Research Proposal. (10 Marks)

QUESTION THREE (20 MARKS)

a) Discuss Research Ethics

(10 Marks)

b) What are some of some of the unethical practices that you may encounter while doing Research? How can you avoid the falling prey to these practices? (10 Marks)

QUESTION FOUR (20 MARKS)

a) Why is a review of literature important in undertaking scientific research?

(15 Marks)

b) What is meant by phrase “SMART Objectives”

(5 Marks)

QUESTION FIVE (20 MARKS)

Using specific examples in computing and Information Technology research, discuss the following research approaches;

- a) Survey Research
- b) Experimental Research
- c) Descriptive research
- d) Exploratory Research

(20 Marks)